

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I **Devadharshini E (192524483)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Devadharshini E

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility**Knowledge Base**

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System**Knowledge Base**

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

QUESTION 1 – RAIN AND WET GROUND

1. Propositional Logic Representation

Let:

R = It is raining

W = The ground is wet

The first statement says that whenever it rains, the ground becomes wet. This can be written as:

$R \rightarrow W$

The second statement tells us that it is currently raining: R

The goal is to prove: W

2. Conversion into CNF

For the first statement: $R \rightarrow W$

After removing the implication: $\neg R \vee W$

So, the CNF form is: $\neg R \vee W$

The second statement is: R

It is already in CNF form.

Now, we negate the goal: W

Negated goal: $\neg W$

Therefore, the clauses used for resolution are:

1. $\neg R \vee W$
2. R
3. $\neg W$

3. Resolution Algorithm

Step 1:

From: $\neg R \vee W$ and R, R and $\neg R$ cancel each other.

Therefore: W

Step 2:

Now we have: W and $\neg W$, W and $\neg W$ contradict each other.

Therefore: $\square$

4. Final Result

The resolution process produces the empty clause $\square$. This means that the negation of the goal leads to a contradiction.

Therefore, the goal is proved:

The ground is wet.

QUESTION 2 – STUDENT ASSIGNMENT SUBMISSION

1. Propositional Logic Representation

Let:

S = Rahul submitted the assignment

M = Rahul receives internal marks

The first statement says: $S \rightarrow M$

The second statement says: S

The goal is: M

2. Conversion into CNF

For the first statement: $S \rightarrow M$

After removing the implication: $\neg S \vee M$

Therefore, the CNF form is: $\neg S \vee M$

The second statement is: S

This is already in CNF form.

Negating the goal: M

Negated goal: $\neg M$

Therefore, the clauses are:

1. $\neg S \vee M$
2. S
3. $\neg M$

3. Resolution Algorithm

Step 1:

$\neg S \vee M, S$

Resolving S and $\neg S$: M

Step 2:

M, $\neg M$

Resolving M and $\neg M$:

□

4. Conclusion

The resolution process produces the empty clause □, so the negated goal leads to a contradiction.

Therefore, the goal is successfully proved:

Rahul receives internal marks.

QUESTION 3 – LIBRARY MEMBERSHIP

1. Propositional Logic Representation

Let:

L = Priya is a library member

B = Priya can borrow books

The first statement says:

$L \rightarrow B$

The second statement says: L

The goal is: B

2. Conversion into CNF

For the first statement: $L \rightarrow B$

After removing the implication: $\neg L \vee B$

Therefore, the CNF form is: $\neg L \vee B$

The second statement is: L

This is already in CNF form.

Now, negate the goal: B

Negated goal: $\neg B$

Therefore, the clauses are:

1. $\neg L \vee B$
2. L
3. $\neg B$

3. Resolution

Step 1:

$\neg L \vee B, L$

Resolving L and $\neg L$: B

Step 2:

B, $\neg B$

Resolving B and $\neg B$:

□

No further clauses are needed because the empty clause has been obtained.

4. Final Conclusion

The resolution process produces the empty clause □, which proves that the negated goal is contradictory.

Therefore, **Priya can borrow books.**

QUESTION 4 – PLACEMENT ELIGIBILITY

1. Propositional Logic Representation

Let:

A = Arun cleared the aptitude test

E = Arun is eligible for placement

The first statement says: $A \rightarrow E$

The second statement says: A

The goal is: E

2. CNF Conversion

For the first statement: $A \rightarrow E$

After removing the implication: $\neg A \vee E$

Therefore, the CNF clause is: $\neg A \vee E$

The second statement is: A

This is already in CNF form.

3. Negation of the Goal

Goal: E

Negated goal: $\neg E$

Therefore, the clauses are:

1. $\neg A \vee E$

2. A

3. $\neg E$

4. Resolution Algorithm

Step 1:

$\neg A \vee E$, A

Resolving A and $\neg A$: E

Step 2:

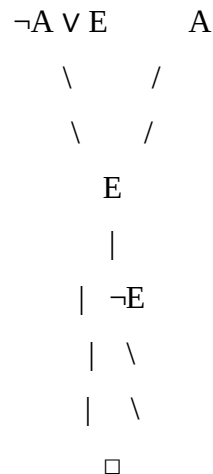
E, $\neg E$

Resolving E and $\neg E$:

□

The empty clause □ is obtained, so the negated goal leads to a contradiction.

5. Resolution Tree



6. Conclusion

Since the resolution process produces the empty clause $\square$, the goal is successfully proved.

Therefore, Arun is eligible for placement.

QUESTION 5 – ACCESS CONTROL SYSTEM

1. Propositional Logic Representation

Let:

P = The user entered the correct password

A = The user is authenticated

G = The user is granted access

The first statement says: $P \rightarrow A$

The second statement says: $A \rightarrow G$

The third statement says: P

The goal is: G

2. Conversion into CNF

For the first implication: $P \rightarrow A$

After removing the implication: $\neg P \vee A$

For the second implication: $A \rightarrow G$

After removing the implication: $\neg A \vee G$

The third statement is: P

This is already in CNF form.

3. Negation of the Goal

Goal: G

Negated goal: $\neg G$

Therefore, the clauses are:

1. $\neg P \vee A$
2. $\neg A \vee G$
3. P
4. $\neg G$

4. Resolution Algorithm

Step 1:

$\neg P \vee A, P$

Resolving P and $\neg P$: A

Step 2:

$\neg A \vee G, A$

Resolving A and $\neg A$: G

Step 3:

$G, \neg G$

Resolving G and $\neg G$:

$\square$

5. Final Conclusion

The resolution process produces the empty clause $\square$, which means that the negated goal leads to a contradiction.

Therefore, the goal is successfully proved:

The user is granted access.